

***Generation Interconnection
System Impact Study Report***

For

***PJM Generation Interconnection Request
Queue Position AB2-177***

“Kenney 25 kV”

May 2017

Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances an Interconnection Customer may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement. The possibility of sharing the reinforcement costs with other projects may be identified in the Feasibility Study, but the actual allocation will be deferred until the System Impact Study is performed.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The Interconnection Customer is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

General

Harron, LLC, the Interconnection Customer (IC), has proposed a 6.6 MW MFO (2.5 MWC) solar generating facility to be located in Snow Hill, Worcester County, Maryland. PJM studied AB2-177 as a 6.6 MW injection into the Delmarva Power and Light Company's (DPL) system at the Kenney 25 kV Substation and evaluated it for compliance with reliability criteria for summer peak conditions in 2020. The planned in-service date, as requested by the IC during the project kick-off call, is December 1, 2018. This date is not attainable due to additional required PJM studies and Transmission Owner construction schedules.

Point of Interconnection

The Interconnection Customer requested a distribution level interconnection. As a result, AB2-177 will interconnect with the Delmarva Power and Light distribution system as follows (see Attachment 1):

The first 5.7 MWs will connect to the existing 25 kV feeder MD 2239 from the 69/25 kV T2 transformer at the Kenney Substation; the next 0.9 MW will connect to the 69/25 kV T2 transformer at the Kenney Substation via an express feeder.

Direct Connection Requirements

Criteria Limits for Distributed Energy Resource (DER) Connections to the ACE, DPL and Pepco Distribution Systems (less than 69kV)

1. Single Phase Limit

Any DER with a capacity that exceeds 100kW shall be a balanced 3 phase system.

2. Voltage Limits

DER's are permitted to cause a voltage fluctuation of up to 2% at the Point of Interconnection, ½ the band width of any voltage regulator at its terminals, and ½ the net dead band of a switched capacitor bank at its connection point. When a DER is at maximum output, it shall not raise the feeder voltage above the ANSI C84.1 or state limit, whichever is more conservative.

3. Existing Distribution Circuit Capacity Limits

The aggregate limit of large (250 kW and over) generators running in parallel with a single, existing distribution circuit is 0.5 MWs on the 4kV, 3MWs on the 12 kV, 6 MWs on the 25 kV, and 10 MWs on the 34 kV.

4. Express Circuit Capacity Limits

Distributed generation installations which exceed the limit for an existing circuit require an express circuit.

The maximum generator size for express circuits shall be:

- 4 kV 0.5 MW
- 12 – 13.8 kV 10 MW
- 23 – 25 kV 10 MW
- 33.26 – 34.5 kV 15 MW

5. Distribution Power Transformer Limit

The aggregate limit of large (250 kW and over) generator injection to a single distribution transformer of 22.5 MVA nameplate or larger is 10 MWs. Transformers with nameplate ratings lower than 22.5 MVA will be given lower ratings on an individual basis. If the transformer rating is significantly greater than 40 MVA it may be possible to consider a greater generation capacity.

Adding a new transformer will be considered if there is no availability on any of the existing transformers and space is available in an existing substation. Any proposed transformers would be PHI's standard distribution transformer.

6. Express Circuit Length Limit

If there is no space for an additional transformer at the closest substation, the next closest substation will be considered. The length of an express circuit is limited to 5 miles, or for the sake of the feasibility study, 3.8 straight line miles to the substation. This simplification is used because the feasibility study phase does not allow for the time and resources to examine routes in detail (including existing pole lines, easements, ROW, and environmental issues etc.)

7. When a New Substation is Required

If a distribution express circuit can't be built from an existing substation for a project, it will be necessary to construct a new distribution substation with a standard ring bus design. It will be supplied by extending existing transmission lines. It is the developer's responsibility to verify eligibility of this configuration for solar renewable energy certificates.

All limits, given above in MWs, are subject to more detailed study to ensure feasibility.

Transmission Owner Scope of Work

TO work required to accommodate 5.7 MW of generation on existing feeder MD2239 from Kenney Substation T2:

1. A utility operated recloser equipped with the proper relaying and communications will be required.
2. Utility grade primary metering will be required.
3. Generation telemetry and remote trip capability will be provided to the control center.
4. A detailed, time-based study may be performed during later study phases.
5. Protection, Planning, and other engineering departments will perform studies, design work, and prepare engineering estimates.
6. Transfer trip will be required. The cost included below is to install 48SM ADSS fiber optic cable in conduit from Kenney Substation to the POI, a total distance of approximately 3.5 miles. 69 kV potential transformers will need to be installed if none already in-service.
7. The feeder will need to be extended approximately 0.6 miles.
8. If the voltage regulators in the substation are not properly equipped to handle the reverse power flow that the generator will cause, adjustments to their controllers will be required. If regulator cannot operate with controller, entire regulator will be upgraded.

Estimated Costs			
Keeney Substation T2			
Fiber Installation	3.5	miles	\$175,000
Feeder Extension	0.6	miles	\$240,000
substation Relaying & 69kV PTs			\$200,000
Recloser & Metering			\$80,000
SCADA Integration into EMS			\$10,000
Dynamic Study			\$30,000
Various Departments Work			\$60,000
Subtotal Cost			\$795,000
Approximate Total Cost with 18% Contingency			\$914,250

The estimated time to complete this work is **18-24 months** after receipt of a fully executed interconnection agreement.

TO work required to accommodate 0.9MW of generation on an express feeder from Kenney Substation T2:

1. Design and construct one new 25 kV feeder with 477 aluminum conductor from Kenney Substation to the generation site – approximately 3.5 miles.
2. One new 25 kV feeder terminal position will be constructed.

3. A utility operated recloser equipped with the proper relaying and communications will be required.
4. Utility grade primary metering will be required for each feeder.
5. Generation telemetry and remote trip capability will be provided to the control center.
6. A detailed, time-based study may be performed during later study phases.
7. Protection, Planning, and other engineering departments will perform studies, design work, and prepare engineering estimates.
8. Transfer trip will be required. The cost included below is to install 48SM ADSS fiber optic cable in conduit from Kenney Substation to the POI, a total distance of approximately 3.5 miles. 69 kV potential transformers will need to be installed if none already in-service.
9. If the voltage regulators in the substation are not properly equipped to handle the reverse power flow that the generator will cause, adjustments to their controllers will be required. If regulator cannot operate with controller, entire regulator will be upgraded.

Estimated Costs			
Keeney Substation T2			
477 AAC Express Feeder	3.5	Miles	\$1,400,000
Fiber Installation			\$175,000
New Feeder Terminal			\$300,000
Substation Relayiong & 69kV PTs			\$200,000
Recloser & Metering			\$80,000
SCADA Integration into EMS			\$10,000
Dynamic Study			\$30,000
Various Departments Work			\$90,000
Subtotal Cost			\$2,285,000
Approximate Total Cost with 18% Contingency			\$2,627,750

The estimated time to complete this work is **36-48 months** after receipt of a fully executed interconnection agreement.

Interconnection Customer Scope of Work

The Interconnection Customer (IC) is responsible for all design and construction related to activities on their side of the point of Interconnection. Site preparation, including grading and an access road, as necessary, is assumed to be by the IC. Route selection, line design, and right-of-way acquisition of the direct connect facilities is not included in this report, and is the responsibility of the IC.

Protective relaying and metering design and installation must comply with PHI's applicable standards. The IC is also required to provide revenue metering and real-time telemetering data to PJM in conformance with the requirements contained in PJM Manuals M-01 and M-14 and the PJM Tariff.

Equipment Requirements

Any transformers on the IC's side must be Wye grounded on the utility side or alternatively 3 phase potential transformers and a relay capable of detecting over/under voltage shall be installed to detect an undesirable condition on the high side of the IC's transformer.

The inverter at the DG location shall have the following capabilities:

- Voltage flicker reduction through dynamic VAR or fixed PF response
- Ramp rate control
- SCADA communications
- Curtailment or other mitigation ability if high voltage were to occur
- Low voltage and system disturbance ride through
- Ability to receive and respond to a transfer trip signal
- Ability to adjust PF or VARs based on utility signal
- Ability to Adjust Real Power Output based on utility signal

The inverter shall operate in accordance with the IEEE 1547 series of standards that have been approved. While inverters should be capable of voltage stabilization thru dynamic VAR response and capable of low voltage and system disturbance ride through, neither of these capabilities shall be implemented until such time that the IEEE 1547 series of standards are revised and approved to include standards for these capabilities. At such time as these revised standards become available, the PV owner/operator shall cooperate with the Company (the 'Company' referring to ACE, DPL, or PEPCO) to implement these capabilities with settings acceptable to the Company. The inverters shall be capable of operating at a fixed power factor value between 0.95 lead and 0.95 lag as specified by the Company. The value is supplied below:

1. Operate inverters at 1.00 unity power factor

Note: In the future, the company reserves the right to issue new fixed power factor setting requirements (0.95 lead to 0.95 lag) if necessary.

It is the responsibility of the owner to secure the inverter from any unauthorized access (including physical and remote access) which could alter settings or adversely affect the inverter's ability to operate as required. Security measures should include utilizing secure password settings and/or physical locks on cabinet doors.

Additional Operating Requirements

1. The Company (DPL, ACE, Pepco) will require the capability to remotely disconnect the generator from the grid by communication from its System Operations facility. This will be accomplished with a line recloser.
2. It is the Interconnection Customer's responsibility to send the data that PJM and the Company requires directly to PJM. The Interconnection Customer will grant permission for PJM to send the Company the following telemetry that the Interconnection Customer sends to PJM: real time MW, MVAR, volts, amperes, generator breaker status or inverter status, and interval MWH and MVARH.

3. The Interconnection Customer will be required to make provisions for a voice quality phone line within approximately 3 feet of each Company metering position to facilitate remote interrogation and data collection.
4. A mutually acceptable means of interrupting and disconnecting the generator with a visible break, able to be tagged and locked out, shall be worked out with Company Distribution Engineering.
5. Company reserves the right to charge the Interconnection Customer operation and maintenance expenses to maintain the Interconnection Customer attachment facilities, including metering and telecommunications facilities, owned by Company.

High Voltage Warning

Voltage received at the meter from the utility can be 104% or 105% of nominal. Normal operating procedures dictate that voltage at the substation be raised to the higher end of an acceptable bandwidth in order to provide adequate supply to distant customers. Transformers with no load taps should be used to reduce the voltage by 2.5% to avoid the possibility of inverter trips. Failure to account for this may result in lost energy production.

Summer Peak Analysis - 2020

Transmission Network Impacts

Potential transmission network impacts are as follows:

Generator Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

None

Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

1. (DP&L - DP&L) The LORET_69-FRUITLND 69 kV line (from bus 232275 to bus 232288 ckt 1) loads from 99.77% to 100.14% (AC power flow) of its emergency rating (137 MVA) for the line fault with failed breaker contingency outage of 'DP56'. This project contributes approximately 0.61 MW to the thermal violation.

CONTINGENCY 'DP56'/*LORETTO BUS BREAKER
 DISCONNECT BRANCH FROM BUS 232127 TO BUS 232117 CKT 1/*LORETTO VIENNA
 138 1380
 DISCONNECT BRANCH FROM BUS 232127 TO BUS 232128 CKT 1/*LORETTO PINEY
 GROVE 138 138
 END

Please refer to Appendix 1 for a table containing the generators having contribution to this flowgate.

Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

1. (DP&L - DP&L) The PINEY_69-M HERMON 69 kV line (from bus 232274 to bus 232272 ckt 1) loads from 105.62% to 106.91% (AC power flow) of its emergency rating (174 MVA) for the line fault with failed breaker contingency outage of 'DP15'. This project contributes approximately 2.16 MW to the thermal violation.

CONTINGENCY 'DP15'/*INDIAN RIVER BUS BREAKER TO PINEY GROVE
DISCONNECT BRANCH FROM BUS 232007 TO BUS 232006 CKT 1/*PINEY GR INDRIV 4
230 230
DISCONNECT BRANCH FROM BUS 232007 TO BUS 232128 CKT 1/*PINEY GR PINEY
GR 230 138
DISCONNECT BRANCH FROM BUS 232006 TO BUS 232004 CKT 1/*MILFORD INDIAN
RIVER 230 230
END

Please refer to Appendix 2 for a table containing the generators having contribution to this flowgate.

Summer Peak Load Flow Analysis Reinforcements

New System Reinforcements

(Upgrades required to mitigate reliability criteria violations, i.e. Network Impacts, initially caused by the addition of this project generation)

1. To mitigate the (DP&L - DP&L) LORETTO 69-FRUITLAND 69 kV line (from bus 232275 to bus 232288 ckt 1) overload will require the rebuild of the Loretto – Fruitland 69 kV transmission line and substation reinforcements at Loretto Substation and Fruitland Substation. The estimated costs to perform this work is **\$7,196,000** and will take approximately **3 years** to complete.

Contribution to Previously Identified System Reinforcements

(Overloads initially caused by prior Queue positions with additional contribution to overloading by this project. This project may have a % allocation cost responsibility which will be calculated and reported for the Impact Study)

1. To mitigate the (DP&L - DP&L) PINEY_69-M HERMON 69 kV line (from bus 232274 to bus 232272 ckt 1) overload include the rebuild of the Piney Grove – Mount Hermon 69 kV transmission line and substation reinforcements at Piney Grove Substation and Mount Hermon Substation. The estimated cost to perform this work is **\$9,688,000** and will take **approximately 3 years** to complete.

Note: AB2-177 will not have any cost allocation toward this upgrade at this time because of percent MW contribution. However, if prior projects withdraw, such that the AB2-177 project now becomes “first to cause”, there may be cost allocation.

Steady-State Voltage Requirements

No issues identified.

Short Circuit

Not required.

Stability and Reactive Power Requirement

Not required.

Delivery of Energy Portion of Interconnection Request

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Only the most severely overloaded conditions are listed. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed, which will study all overload conditions associated with the overloaded element(s) identified.

1. (DP&L - DP&L) The MILF_230-STEELE 230 kV line (from bus 232004 to bus 232000 ckt 1) loads from 99.8% to 100.06% (AC power flow) of its emergency rating (551 MVA) for the single line contingency outage of 'CKT 23032'. This project contributes approximately 1.67 MW to the thermal violation.

CONTINGENCY 'CKT 23032'

OPEN LINE FROM BUS 231004 TO BUS 232003 CIRCUIT 1/RED LION - CARTANZA 230
END

2. (DP&L - DP&L) The PINEY138-LORETTO 138 kV line (from bus 232128 to bus 232127 ckt 1) loads from 111.63% to 112.74% (AC power flow) of its emergency rating (159 MVA) for the single line contingency outage of 'CKT 13713'. This project contributes approximately 1.8 MW to the thermal violation.

CONTINGENCY 'CKT 13713'

OPEN LINE FROM BUS 232129 TO BUS 232127 CIRCUIT 1/KINGS CREEK - LORETTO
138
END

3. (DP&L - DP&L) The PINEY_69-M HERMON 69 kV line (from bus 232274 to bus 232272 ckt 1) loads from 105.26% to 106.55% (AC power flow) of its emergency rating (174 MVA) for the single line contingency outage of 'CKT 23002'. This project contributes approximately 2.17 MW to the thermal violation.

CONTINGENCY 'CKT 23002'

DISCONNECT BUS 232007/INDIAN RIVER - PINEY GROVE 230 & PNY GRV AT-20

XFMER

END

4. (DP&L - DP&L) The SHORT 1-LAUREL 69 kV line (from bus 232828 to bus 232249 ckt 1) loads from 97.21% to 98.36% (AC power flow) of its emergency rating (57 MVA) for the single line contingency outage of 'CKT 23002'. This project contributes approximately 0.73 MW to the thermal violation.

CONTINGENCY 'CKT 23002'

DISCONNECT BUS 232007/INDIAN RIVER - PINEY GROVE 230 & PNY GRV AT-20

XFMER

END

Light Load Analysis - 2020

Light Load Studies to be conducted during later study phases (as required by PJM Manual 14B).

Facilities Study Estimate

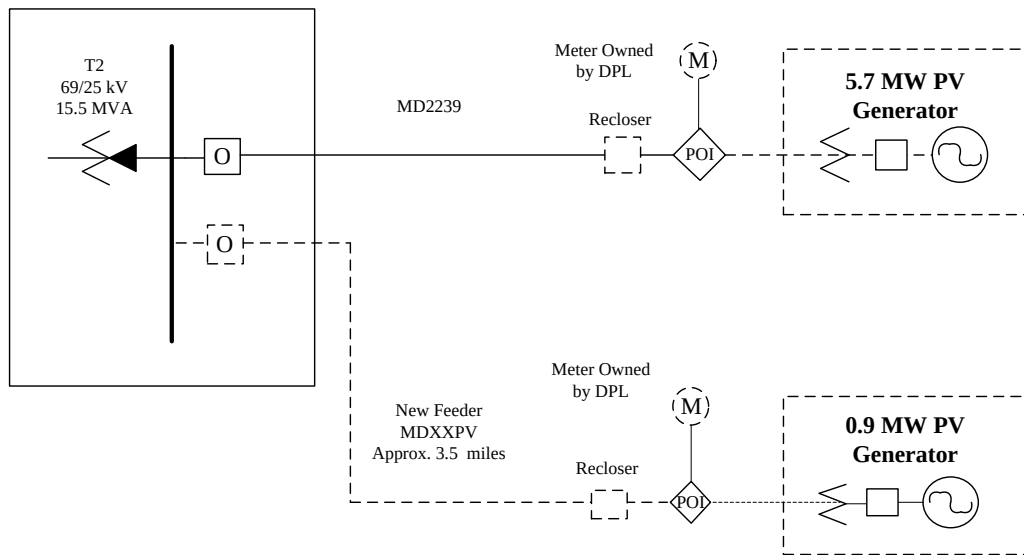
The estimated time for PJM to issue a Facilities Study Report is 7 months. The deposit required for project will be \$50,000.

AB2-177

Kenney 69/25 kV Sub

6.6 MW PV Solar Generator

Kenney Substation



Appendices

The following appendices contain additional information about each flowgate presented in the body of the report. For each appendix, a description of the flowgate and its contingency was included for convenience. However, the intent of the appendix section is to provide more information on which projects/generators have contributions to the flowgate in question. Although this information is not used "as is" for cost allocation purposes, it can be used to gauge other generators impact.

It should be noted the generator contributions presented in the appendices sections are full contributions, whereas in the body of the report, those contributions take into consideration the commercial probability of each project.

Appendix 1

(DP&L - DP&L) The LORET_69-FRUITLND 69 kV line (from bus 232275 to bus 232288 ckt 1) loads from 99.77% to 100.14% (AC power flow) of its emergency rating (137 MVA) for the line fault with failed breaker contingency outage of 'DP56'. This project contributes approximately 0.61 MW to the thermal violation.

CONTINGENCY 'DP56' /*LORETTO BUS BREAKER
DISCONNECT BRANCH FROM BUS 232127 TO BUS 232117 CKT 1 /*LORETTO
VIENNA 138 1380
DISCONNECT BRANCH FROM BUS 232127 TO BUS 232128 CKT 1 /*LORETTO
PINEY GROVE 138 138
END

<i>Bus Number</i>	<i>Bus Name</i>	<i>Full Contribution</i>
232905	BAYVIEW1	0.44
232926	CRISFLD1	0.67
232912	OH NUG1	1.6
232913	OH NUG2	1.58
232914	OH NUG3	1.6
232915	OH NUG4	1.6
232916	OH NUG5	1.6
232917	OH NUG6	1.59
232918	OH NUG7	1.59
232921	TASLEY2G	1.1
904210	V4-022C	0.07
904212	V4-022E	0.54
901003	W1-003 C	0.28
901004	W1-003 E	0.82
901013	W1-004 C	0.28
901014	W1-004 E	0.82
901023	W1-005 C	0.28
901024	W1-005 E	0.82

901033	W1-006 C	0.28
901034	W1-006 E	0.82
907052	X1-032 E	1.1
907323	X1-096 C	1.31
907324	X1-096 E	30.34
920582	Z1-076 C	0.67
920583	Z1-076 E	1.09
920592	Z1-077 C	0.48
920593	Z1-077 E	0.78
916441	Z1-100	0.15
916451	Z1-101	0.15
916461	Z1-102	0.15
920602	Z1-103	0.15
917081	Z2-012 C	0.27
917082	Z2-012 E	2.15
920952	AA1-025	0.14
920962	AA1-026	0.14
920972	AA1-027	0.14
920982	AA1-028	0.14
921122	AA1-059 C	0.29
921123	AA1-059 E	0.55
918831	AA1-102	2.51
921602	AA1-141 C	0.52
921603	AA1-141 E	0.85
922213	AA2-129 E	3.46
922222	AA2-130	0.65
923902	AB2-030 E	0.69
923931	AB2-033 C	1.24
923932	AB2-033 E	0.49
924361	AB2-084 C	1.04
924362	AB2-084 E	1.7
924681	AB2-120 C OP	6.05
924682	AB2-120 E OP	9.88
925101	AB2-167 C	0.66
925102	AB2-167 E	1.09
925231	AB2-177 C	0.23
925232	AB2-177 E	0.38

Appendix 2

(DP&L - DP&L) The PINEY_69-M HERMON 69 kV line (from bus 232274 to bus 232272 ckt 1) loads from 105.62% to 106.91% (AC power flow) of its emergency rating (174 MVA) for the line fault with failed breaker contingency outage of 'DP15'. This project contributes approximately 2.16 MW to the thermal violation.

CONTINGENCY 'DP15' /*INDIAN RIVER BUS BREAKER TO
PINEY GROVE

DISCONNECT BRANCH FROM BUS 232007 TO BUS 232006 CKT 1 /*PINEY GR
INDRIV 4 230 230

DISCONNECT BRANCH FROM BUS 232007 TO BUS 232128 CKT 1 /*PINEY GR
PINEY GR 230 138

DISCONNECT BRANCH FROM BUS 232006 TO BUS 232004 CKT 1 /*MILFORD
INDIAN RIVER 230 230

END

<i>Bus Number</i>	<i>Bus Name</i>	<i>Full Contribution</i>
232905	BAYVIEW1	0.62
232926	CRISFLD1	0.38
232912	OH NUG1	2.2
232913	OH NUG2	2.17
232914	OH NUG3	2.2
232915	OH NUG4	2.2
232916	OH NUG5	2.2
232917	OH NUG6	2.19
232918	OH NUG7	2.19
232921	TASLEY2G	1.53
904210	V4-022C	0.09
904212	V4-022E	0.75
901003	W1-003 C	0.37
901004	W1-003 E	1.07
901013	W1-004 C	0.37
901014	W1-004 E	1.07
901023	W1-005 C	0.37
901024	W1-005 E	1.07
901033	W1-006 C	0.37
901034	W1-006 E	1.07
907052	X1-032 E	0.82
907323	X1-096 C	0.75
907324	X1-096 E	17.31
920582	Z1-076 C	1.55
920583	Z1-076 E	2.52
920592	Z1-077 C	1.1
920593	Z1-077 E	1.8
916441	Z1-100	0.2

916451	Z1-101	0.2
916461	Z1-102	0.2
920602	Z1-103	0.2
917081	Z2-012 C	0.38
917082	Z2-012 E	2.99
920952	AA1-025	0.18
920962	AA1-026	0.18
920972	AA1-027	0.18
920982	AA1-028	0.18
921122	AA1-059 C	0.16
921123	AA1-059 E	0.32
918831	AA1-102	1.43
921602	AA1-141 C	1.86
921603	AA1-141 E	3.04
922213	AA2-129 E	4.76
922222	AA2-130	0.37
923902	AB2-030 E	0.97
923931	AB2-033 C	1.73
923932	AB2-033 E	0.68
924361	AB2-084 C	0.78
924362	AB2-084 E	1.27
924681	AB2-120 C OP	9.17
924682	AB2-120 E OP	14.96
925101	AB2-167 C	1.54
925102	AB2-167 E	2.53
925231	AB2-177 C	0.82
925232	AB2-177 E	1.34